

Embedded systems exp 15

Program:

```
#include <reg51.h>
```

```
void main(void)
```

```
{
```

```
    unsigned char hexa = 0xFF;
```

```
    unsigned char hundreds, tens, units;
```

```
    ACC = hexa;
```

```
    B = 10;
```

```
    ACC = ACC / B;
```

```
    units = B;
```

```
    B = 10;
```

```
    ACC = ACC / B;
```

```
    tens = B;
```

```
    hundreds = ACC;
```

```
    P0 = units;
```

```
    P1 = tens;
```

```
    P2 = hundreds;
```

```
    while(1);
```

```
}
```

Output:

C:\Keil_v5\C51\Examples\HELLO\keil\exp15.uvproj - µVision

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Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x05

Disassembly

```
14: ACC = ACC / B;
15: tens = B;
16:
```

exp15.c STARTUP.A51

```
8 ACC = hexa;
9 B = 10;
10 ACC = ACC / B;
11 units = B;
12
13 B = 10;
14 ACC = ACC / B;
15 tens = B;
16
17 hundreds = ACC;
18
19 P0 = units;
20 P1 = tens;
21 P2 = hundreds;
22
23 while(1);
24 }
```

Parallel Port 1

Port 1

P1: 0x05

Pins: 0x05

Parallel Port 0

Port 0

P0: 0x05

Pins: 0x05

Parallel Port 2

Port 2

P2: 0x02

Pins: 0x02

Command

Running with Code Size Limit: 2K

Load "C:\Keil_v5\C51\Examples\HELLO\keil\Objects\exp15"

BS \EXP15\14, 1

BS \?C_STARTUP\135, 1

ASH ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess

Call Stack + Locals

Na...	Location/Value	Type
	C:0x080B	
	0x05	uchar
	0x00	uchar
	0x00	uchar
	0x05	uchar